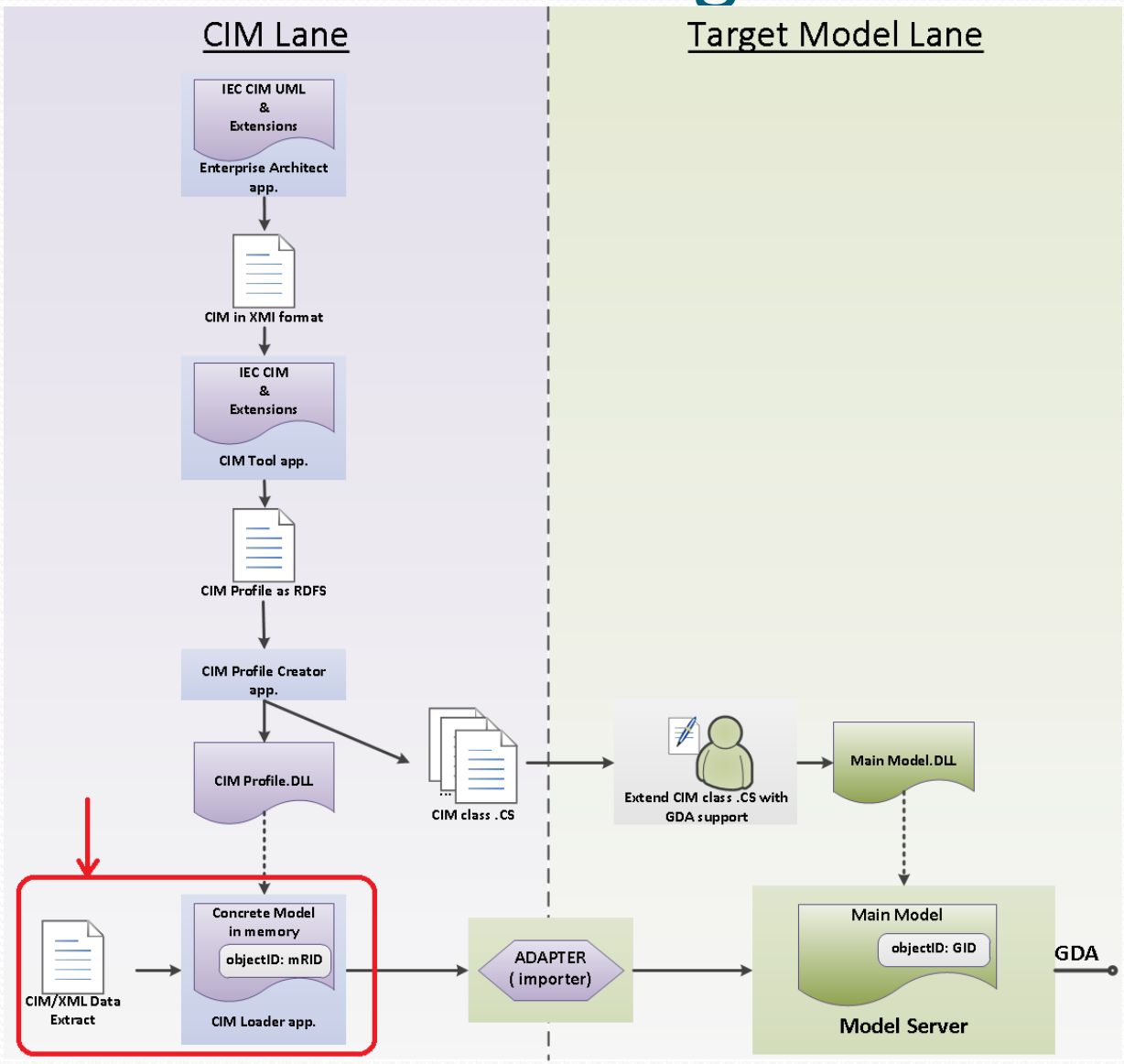
Standardi i modeliranje elektroenergetskih sistema

VEŽBA 4b:

Model loader

Tok podataka pri inicijalizaciji modela elektroenergetske mreže



Model loader 1/2

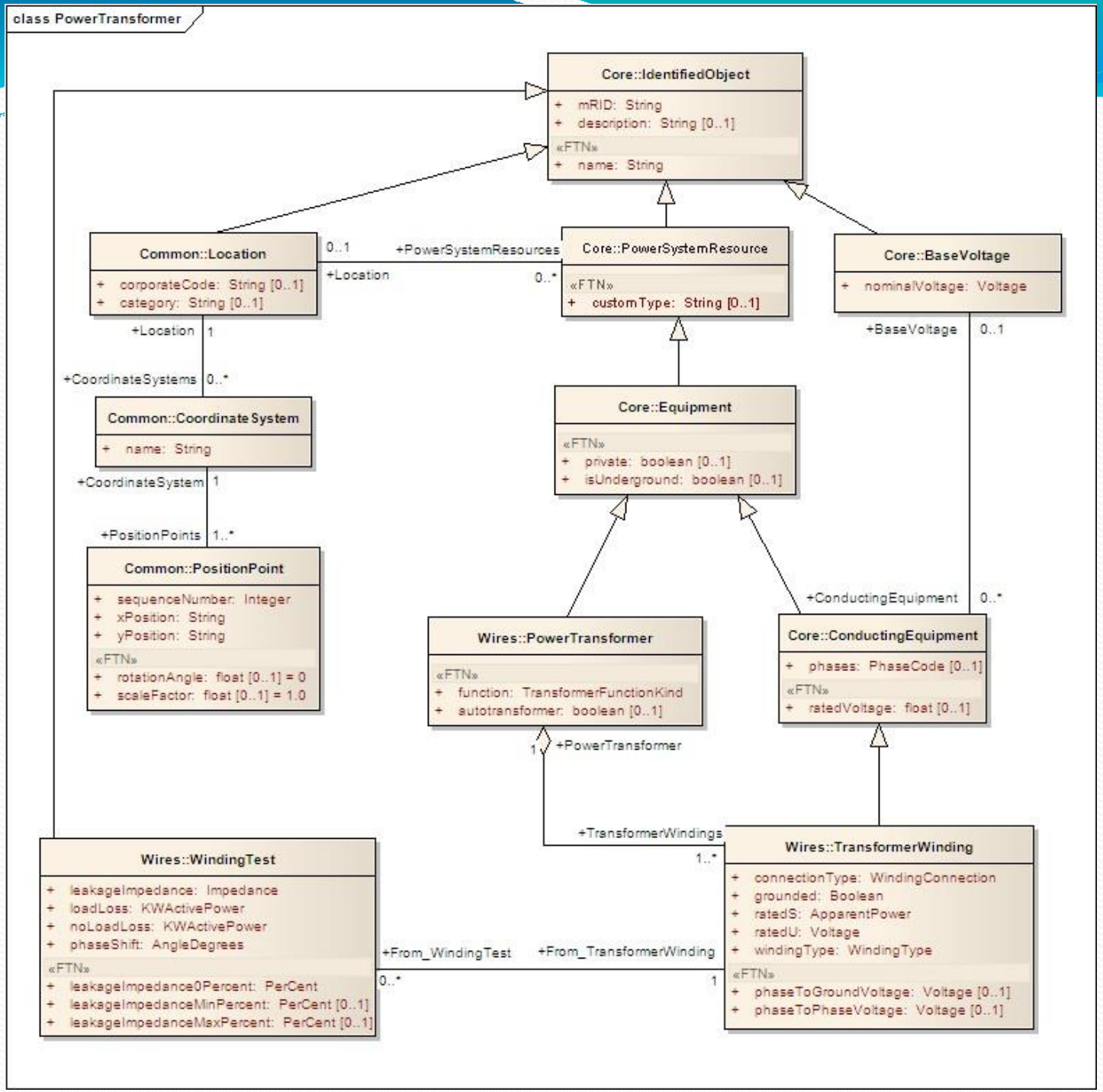
- ConcreteModel klasa
  - Model podataka koji se kreira na osnovu CIM/XML-a
  - Sadrži mapu svih entiteta
  - Obezbeđuje metode za upis i čitanje entiteta

Model loader 2/2

- … ConcreteModel klasa
  - Organizacija entiteta po tipu
  - SortedDictionary<string, SortedDictionary<string, object>> modelMap
  - “Spolja” mapa kao ključ ima tip entiteta, dok je vrednost nova (“unutrašnja”) mapa koja sadrži entitete istog tipa
  - “Unutrašnja” mapa kao ključ ima mrID entiteta, dok je vrednost sam entitet

Zadaci

1. Dodati ispis broja entiteta svakog tipa
2. Ispisati vrednost atributa *LoadLoss* entiteta tipa *WindingTest* čiji je *mrID* jednak *"939140759_TW1_WT"*.


class PowerTransformer
Core::IdentifiedObject
+ mRID: String
+ description: String [0..1]
«FTN»
+ name: String
Common::Location
+ corporateCode: String [0..1]
+ category: String [0..1]
0..1
+PowerSystemResources
+Location
0..*
Core::PowerSystemResource
«FTN»
+ customType: String [0..1]
Core::BaseVoltage
+ nominalVoltage: Voltage
+Location
1
+BaseVoltage
0..1
+CoordinateSystems
0..*
Common::CoordinateSystem
+ name: String
Core::Equipment
«FTN»
+ private: boolean [0..1]
+ isUnderground: boolean [0..1]
+CoordinateSystem
1
+PositionPoints
1..*
Common::PositionPoint
+ sequenceNumber: Integer
+ xPosition: String
+ yPosition: String
«FTN»
+ rotationAngle: float [0..1] = 0
+ scaleFactor: float [0..1] = 1.0
+ConductingEquipment
0..*
Wires::PowerTransformer
«FTN»
+ function: TransformerFunctionKind
+ autotransformer: boolean [0..1]
Core::ConductingEquipment
+ phases: PhaseCode [0..1]
«FTN»
+ ratedVoltage: float [0..1]
1
+PowerTransformer
+TransformerWindings
1..*
Wires::WindingTest
+ leakageImpedance: Impedance
+ loadLoss: KWActivePower
+ noLoadLoss: KWActivePower
+ phaseShift: AngleDegrees
«FTN»
+ leakageImpedance0Percent: PerCent
+ leakageImpedanceMinPercent: PerCent [0..1]
+ leakageImpedanceMaxPercent: PerCent [0..1]
+From_WindingTest
+From_TransformerWinding
0..*
1
Wires::TransformerWinding
+ connectionType: WindingConnection
+ grounded: Boolean
+ ratedS: ApparentPower
+ ratedU: Voltage
+ windingType: WindingType
«FTN»
+ phaseToGroundVoltage: Voltage [0..1]
+ phaseToPhaseVoltage: Voltage [0..1]